

Face-to-Face Exports: The Role of Business Travel in Trade Promotion

Journal of Travel Research
51(5) 632–639
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DOI: 10.1177/0047287512437857
<http://jtr.sagepub.com>


Maksim Belenkiy¹ and David Riker¹

Abstract

In this study, we investigate the contribution of international business travel to U.S. exports. We present a Dixit-Stiglitz monopolistic competition model that relates business travel to the volume of international trade. Then we estimate the parameters of the model. To avoid the likely bias in OLS estimation, we develop an instrumental variable estimator. We find that lowering the costs of business travel to a country will have a significant positive impact on U.S. commodity exports to the visited country. The elasticity of exports with respect to international business travel is approximately 0.46. Using this estimate, we calculate that each additional international business trip will increase U.S. commodity exports to the visited country by \$36,693 per year on average.

Keywords

business travel, international trade, econometrics

Introduction

International business travelers facilitate trade through face-to-face interactions. They negotiate sales, promote brands, and establish joint ventures. The resulting expansion of a country's exports can generate significant benefits for the exporting economy. According to Helpman, Melitz, and Yeaple (2004), Eaton, Kortum, and Kramarz (2004), and Bernard et al. (2007), exporting firms are significantly more productive, more capital and technology intensive, and more efficient.

The idea that international business travel is a necessary ingredient of international trade has been explored in the travel research literature dating back at least to 1970. Gray (1970) and Keintz (1971) documented a link between international travel and international trade. More recently, Turner and Witt (2001) found that international trade has a significant influence on business travel. An important issue raised by these studies is that international business travel and international trade are jointly determined (or, in more technical jargon, jointly endogenous). This leads to statistical challenges that need to be addressed when quantifying the contribution of business travel to international trade.

A number of studies have used time-series econometric techniques to address the issue of the joint endogeneity of travel and trade. The most common approach is to use Granger causality tests to examine whether changes in the volume of travel precede and predict changes in the volume of trade. Kulendran and Wilson (2000) examine international trade and travel data for Australia and four major trading partners. They test whether business travel leads to trade.

Based on Granger causality tests for the sample period between 1982 and 1997, they conclude that in the case of the United States, business and leisure travel Granger-causes total trade and specifically that travel Granger-causes exports. The authors find similar evidence of Granger causality for Japan and the United Kingdom. In a related study, Shan and Wilson (2001) investigate the time-series relationship between the international tourism and international trade flows of China. Khan, Toh, and Chua (2005) find evidence of cointegration and Granger causality between the tourism and trade of Singapore.

While all of these studies reinforce the need to correct for the joint endogeneity of travel and trade, they address the endogeneity of travel in a particular way: they adopt the identifying assumption that international travel that precedes trade flows is not responding to expectations about the future demand for the exports. However, an alternative interpretation is that precedence simply reflects the lag between face-to-face sales efforts and consequent international shipments, in which case the two variables are jointly determined though temporally mismatched.

Instrumental variables estimation is an alternative to the Granger-causality approach that does not rely on this assumption about timing. Several recent econometric studies of

¹U.S. Department of Commerce, Washington, DC, USA

Corresponding Author:

David Riker, Economist, International Trade Administration,
U.S. Department of Commerce, 1401 Constitution Avenue, N.W.,
Room 2815, Washington, DC 20230
Email: david_riker@verizon.net

travel and trade have adopted this approach, at least in part. In an analysis of firm-level data, Aradhyula and Tronstad (2003) find that business travel increase U.S. exports to Mexico. Poole (2010) finds that there is a positive effect of international business travelers who visit the United States on the volume of U.S. exports, but she finds no significant effect of outbound U.S. business travelers on U.S. exports. This is a surprising result, given the time-series evidence in Kulendran and Wilson (2000) and others. Her surprising result for outbound U.S. business travelers probably reflects a negative bias because of the endogeneity of business travel. Although Poole uses instrumental variables to correct for bias in her model of international business travelers who visit the United States, she does not use instrumental variables to correct for the bias in her model of the outbound U.S. business travelers.

Cristea (2011) uses an instrumental variables strategy that more effectively deals with the joint endogeneity of outbound U.S. business traveler and U.S. exports. She estimates the responsiveness of the demand for international business travel to the scale and composition of U.S. manufacturing exports, conditional on the cost of the travel. She finds that there is a significant relationship between outbound U.S. business travel and U.S. exports, as we would expect. However, because her model estimates the demand for business travel conditional on the level of trade, it cannot be used to quantify the effect on U.S. exports of a change in the cost of international business travel.

In this article, we focus on this practical question: How large of an increase in U.S. commodity exports can be achieved through face-to-face interactions that are facilitated by international business travel? We provide an econometric analysis that further develops the instrumental variables approach but also includes aspects of the Granger causality studies. We incorporate business travel into a model of international trade, following the literature on informative advertising. Following Arkolakis (2008), we assume that firms undertake marketing in order to reach an expanded set of consumers in foreign markets. In the model, the number of business travelers to each country is a choice of each firm, with costs and benefits. We estimate the parameters of the model using data on exports from the United States to 21 countries (or groups of countries) between 1996 and 2009. To correct for potential endogeneity bias, we propose an instrumental variable that is strongly correlated with the number of business travelers but is not likely to have an effect on commodity trade other than through its effect on the number of business travelers. The number of leisure (i.e., nonbusiness) travelers to a country in a given year serves as the instrument for the number of international business travelers in the same year. This assumption is consistent with time series evidence in Kulendran and Witt (2003) that holiday tourism is an important determinant of international business travel.

Model of International Business Travel and Trade

We derive the econometric specification from a Dixit-Stiglitz monopolistic competition model of trade with CES preferences. Krugman (1980) provides one of the first examples of this type of international trade model. We assume that foreign buyers respond positively to the targeted sales efforts of business travelers from the United States. Exporters choose their marketing efforts to maximize their profits from export sales.

In the economic model, there is a continuum of firms in the United States that export. Each firm produces a unique variety. The variable q_j in (1) represents the consumption of each variety in country j in year t . Since we only model business travelers and exports from the United States, we omit subscripts for the country of origin to simplify the notation.

$$q_{jt} = \frac{A(F_{jt}, B_{jt}, G_{jt}) (P_{jt})^{-\sigma} (Y_{jt})^{\alpha}}{(Z_{jt})^{1-\sigma}}. \quad (1)$$

$A(F_{jt}, B_{jt}, G_{jt})$ is a function of several factors that shift the demand for U.S. products in country j : F_{jt} represents a set of export demand factors in country j , including deficiencies in brand awareness and the intensity of foreign competition; B_{jt} measures the targeted marketing efforts of business travelers from the United States; and G_{jt} represents a set of additional country characteristics, including trade barriers and cultural ties with the United States. The partial derivative of A with respect to F_{jt} is negative, and the partial derivative of A with respect to B_{jt} is positive. We assume that consumer demands in country j have a constant elasticity of substitution between the differentiated products, represented by the parameter σ . The variable Y_j is the level of aggregate expenditures in country j . The parameter α is the income elasticity of export demand. To increase the flexibility of the demand function in (1), we do not impose the conventional restriction that α is equal to one. The variables P_{jt} and Z_{jt} are the price of the U.S. varieties in country j and the overall price index in the country in year t .

We assume that there is monopolistic competition among exporters. Given the demand function in (1), their delivered price in country j is equal to a constant mark-up over the marginal cost of the U.S. exporters (represented by M) and the variable cost of shipping the product to country j (represented by S_{jt}). This shipping cost is the ratio of the delivered price in country j to the FAS price in the United States.

$$P_{jt} = \frac{\sigma}{\sigma-1} M_t S_{jt}. \quad (2)$$

By substituting the pricing rule (2) into the demand function in (1), we obtain the following reduced-form expression for the total value of the U.S. commodity exports to country j .

$$X_j = N_t A(F_{jt}, B_{jt}, G_{jt}) \left(\frac{\sigma M_t S_{jt}}{(\sigma-1) Z_{jt}} \right)^{1-\sigma} (Y_{jt})^\alpha \cdot (3)$$

The variable N represents the number of U.S. producers. We assume that U.S. firms produce differentiated products but have identical costs of production. It is straightforward to extend the model to include heterogeneity in firm productivity, following Melitz (2003), Helpman, Melitz, and Rubinstein (2008), and Chaney (2008). In that case, the number of differentiated products that are exported could also vary by destination country.

An exporter can promote its products abroad by traveling in the foreign country. We assume that each firm chooses its number of business travelers in country j (represented by B_{jt}), and incurs a cost for each traveler (represented by V_{jt}), to maximize the export profit function in (4).

$$\Pi_j = \frac{1}{\sigma} A(F_{jt}, B_{jt}, G_{jt}) \left(\frac{\sigma M_t S_{jt}}{(\sigma-1) Z_{jt}} \right)^{1-\sigma} (Y_{jt})^\alpha - V_{jt} B_{jt} \cdot (4)$$

The first-order condition for profit maximization implicitly defines the optimal level of international business travel, B_{jt} .

$$\frac{1}{\sigma} A_B(F_{jt}, B_{jt}, G_{jt}) \left(\frac{\sigma M_t S_{jt}}{(\sigma-1) Z_{jt}} \right)^{1-\sigma} (Y_{jt})^\alpha = V_{jt} \cdot (5)$$

A_B represents the partial derivative of A with respect to B_{jt} .

By totally differentiating (5), we obtain the following expression for the equilibrium percentage change in the volume of business travel, $\hat{B}_{jt}$.

$$\begin{aligned} \hat{B}_{jt} = & \left(\frac{-A_B \alpha}{B_{jt} A_{BB}} \right) \hat{Y}_{jt} + \left(\frac{-F_{jt} A_{BF}}{B_{jt} A_{BB}} \right) \hat{F}_{jt} + \left(\frac{-G_{jt} A_{BG}}{B_{jt} A_{BB}} \right) \hat{G}_{jt} \\ & + \left(\frac{A_B (\sigma-1)}{B_{jt} A_{BB}} \right) \left(\hat{M}_t + \hat{S}_{jt} - \hat{Z}_{jt} \right) + \left(\frac{V_{jt} A_B}{B_{jt} A_{BB}} \right) \hat{V}_{jt} \cdot \end{aligned} (6)$$

By totally differentiating (3), we obtain the following expression for the equilibrium percentage change in the total value of U.S. commodity exports to the country.

$$\begin{aligned} \hat{X}_{jt} = & \left(\frac{A_B B_{jt}}{A} \right) \hat{B}_{jt} + \alpha \hat{Y}_{jt} + \left(\frac{A_F F_{jt}}{A} \right) \hat{F}_{jt} + \\ & \left(\frac{A_G G_{jt}}{A} \right) \hat{G}_{jt} + \hat{N}_t + (1-\sigma) (\hat{M}_t + \hat{S}_{jt} - \hat{Z}_{jt}). \end{aligned} (7)$$

The log-linear approximations in (6) and (7) determine the coefficients in the following system of log-linear equations, with lower-case variables indicating the natural logs of the variables:

$$\begin{aligned} b_{jt} = & \theta_{BY} y_{jt} + \theta_{BF} f_{jt} + \theta_{BG} g_{jt} + \\ & \theta_{BM} m_t + \theta_{BS} s_{jt} + \theta_{BZ} z_{jt} + \theta_{BV} v_{jt}, \end{aligned} (8)$$

$$\begin{aligned} x_{jt} = & \theta_{XB} b_{jt} + \theta_{XY} y_{jt} + \theta_{XF} f_{jt} + \theta_{XG} g_{jt} + \\ & n_t + \theta_{XM} m_t + \theta_{XS} s_{jt} + \theta_{XZ} z_{jt}. \end{aligned} (9)$$

For example, the coefficient θ_{XB} is the elasticity of the value of U.S. exports with respect to the number of business travelers to country j in year t . The system of log-linear equations in (8) and (9) jointly determines the number of business travelers to country j and the value of U.S. exports to the country. The number of business travelers is increasing in income, brand awareness, and the proximity and size of country j , and it is decreasing in the cost of the international travel. Similarly, the value of U.S. exports is increasing in the volume of business travel and country size and is decreasing in the cost of international shipping.

The key empirical implication of the log-linear system of equations in (8) and (9) is that business travel is not an independent determinant of export volume to country j , because exports and the number of business travelers are both determined by the unobservable export demand factors, represented in these equations by the variables f_{jt} and g_{jt} . This joint endogeneity needs to be addressed in the econometric analysis.

In our econometric estimation, we use the specification in (10) to estimate the parameters in (9).

$$x_{jt} = \gamma_0 + \gamma_1 b_{jt} + \gamma_2 y_{jt} + \gamma_3 s_{jt} + \delta_j + \mu_t + u_{jt} \cdot (10)$$

The variables δ_j and μ_t represent country and time fixed effects. The error term of the econometric model, u_{jt} , includes $\theta_{XF} f_{jt}$ and $\theta_{XG} g_{jt}$. OLS estimation of this elasticity based on equation (10), with unobservable f_{jt} and g_{jt} in the error term, will generate a biased estimate of γ_1 as long as θ_{BF} and θ_{BG} are not equal to zero.

To obtain an unbiased estimate of γ_1 , we use an instrumental variable. The instrumental variable should be correlated with the number of international business travelers but should not depend on any other factors that affect the demand for exports, including the variables represented by f_{jt} and g_{jt} . We propose that the annual count of leisure travelers from the United States to country j satisfies these requirements. We expect that the number of leisure travelers is affected by the cost of traveling to the country as well as many other factors that we do not directly observe, including the quality of historical and cultural attractions that are unrelated to commercial opportunities to export to country j . We use the variable l_{jt} to represent the count of leisure travelers. We expect that l_{jt} is positively correlated with v_{jt} , since both types of international travelers reflect the cost of travel from the United States to the country j , but is not correlated with f_{jt} or g_{jt} .

Table 1. List of Regions Included in the Analysis

Africa
Australia
Caribbean
Central America
China
Eastern Europe
France
Germany
Hong Kong
India
Ireland
Italy
Japan
Middle East
Netherlands
South America
South Korea
Spain
Switzerland
Taiwan
United Kingdom

In the first stage of the estimation, we regress the number of business travelers on the number of leisure travelers, using the specification in (11).

$$b_{jt} = \beta_0 + \beta_1 l_{jt} + \beta_2 y_{jt} + \beta_3 s_{jt} + \xi_j + \zeta_t + e_{jt}. \quad (11)$$

The variables ξ_j and ζ_t represent country and year fixed effects in the first-stage specification. Then we substitute the predicted values of b_{jt} into (10). This two-stage procedure is equivalent to instrumental variable (IV) regression where the number of business travelers is instrumented with the number of leisure travelers.

Data on International Travel and Trade

The data on business and leisure travel are from the outbound U.S. travel and tourism statistics published by the Office of Travel and Tourism (OTTI) of the U.S. Department of Commerce. This publication provides a summary of responses to OTTI's annual Survey of International Air Travelers (SIAT). OTTI reports the fraction of business and leisure travelers to 36 regions (defined as either a single country or a set of countries). From these 36 regions, we focus on the 21 regions listed in Table 1, which all have complete data for the period from 1996 to 2009. Canada and Mexico are not included in our analysis. The SIAT only includes international air travelers from the United States. Since Canada and Mexico can be easily reached by land, the volume of business travel to these two countries is undercounted by the air travel survey.

Table 2. Share of U.S. Business Travelers, Leisure Travelers, and Exports by Region in 2009

Region	Share of Total Business Travelers (%)	Share of Total Leisure Travelers (%)	Share of Total U.S. Exports (%)
Africa	5	3	4
Australia	2	2	3
Caribbean	8	19	3
Central America	4	6	3
China	6	4	10
Eastern Europe	4	4	1
France	5	6	4
Germany	7	5	6
Hong Kong	4	2	2
India	4	4	2
Ireland	2	2	1
Italy	4	6	2
Japan	9	4	7
Korea	4	6	4
Middle East	7	2	7
Netherlands	3	8	4
South America	7	2	9
Spain	2	4	1
Switzerland	3	2	2
Taiwan	3	2	3
United Kingdom	10	8	6

To construct the numbers of business and leisure travelers, we multiply the shares of business and leisure travelers for each region and year by the total numbers of travelers by type and year. The resulting panel data set contains 294 region-year observations.

We examine the volume of exports between the United States and the 21 regions for the period from 1996 to 2009. We use bilateral trade data for each country from the U.S. International Trade Commission. We aggregate the FAS value of U.S. exports to the six multi-country regions (Africa, Caribbean, Central America, Eastern Europe, Middle East, and South America) by adding the corresponding value of exports to the countries in these regions. The total trade value of U.S. commodity exports in 2009 was \$651 billion. After aggregating and matching the data to the international travel counts, the total value of U.S. exports from our set of 21 regions is \$554 billion, which is 85% of all U.S. exports in 2009.

Table 2 reports the share of U.S. business travelers, leisure travelers, and commodity exports by region in 2009. The regions that account for the largest shares of U.S. commodity exports—China, Japan, the Middle East, South America, and the United Kingdom—also account for some of the largest shares of business travelers. Overall, the correlation between the number of U.S. business travelers to each region and the value of U.S. commodity exports is 0.495. On the other hand, the link between leisure travelers

Table 3. Ordinary Least Squares Econometric Model of Exports

Variables	Unrestricted Model	Restricted Model
Log of GDP	0.579 (0.064)	Set at 1
Log of trade cost	-4.875 (1.914)	-1.804 (1.841)
Log of business travelers	0.284 (0.042)	0.219 (0.041)
Year fixed effects	Included	Included
Region fixed effects	Included	Included
R-squared	0.9701	0.9934
Number of observations	294	294

Note: Dependent variable is the log of U.S. exports to the region in the year. Robust standard errors are in parentheses.

and U.S. commodity exports is weaker: several of the regions with relatively large shares of leisure travelers do not have relatively large commodity export shares, including the Caribbean region, France, and Italy.

We construct measures of shipping costs and aggregate demand in each region to control for determinants of the volume of international commodity trade in conventional gravity models. We measure the size of the market in each region as the GDP of the country or the combined GDPs of the group of countries. The GDP data are from United Nations national accounts. Following Helpman, Melitz, and Yeaple (2004), we use the CIF value of U.S. imports from the region to the customs value of these imports as a proxy for the cost of shipping exported commodities from the United States to the region in each year. The CIF ratio reflects the international distance and transport and insurance costs per unit of distance. The ratios in the sample range from 1.009 to 1.077.

Econometric Estimates

We estimate the parameters in (10) using the panel of 21 regions for the period from 1996 to 2009. To control for a variety of country characteristics, including barriers to commodity imports, we estimated (10) with country fixed effects. Since the United States is the source country for all of the trade flows in the data set, the year fixed effects in the specification control for the size of the U.S. production base and production costs.

Table 3 reports the OLS estimates of the parameters of the model in (10). We report two versions of the model, one that imposes the restriction that $\gamma_2 = 1$ and one that is unrestricted. We consider this restriction because the empirical trade literature often assumes that the income elasticity of export demand is equal to one.¹ Both of the OLS estimates of γ_1 in Table 3 indicate that international business travel has a positive and statistically significant effect on U.S. commodity exports. Since the data clearly reject the restriction that $\gamma_2 = 1$ in favor of the unrestricted model, we focus our discussion on the unrestricted estimate. The OLS estimate of γ_1 in the unrestricted model implies that a 10% increase in the number

Table 4. First-Stage Estimates of the Number of Business Travelers

Variables	
Log of GDP	0.276 (0.082)
Log of trade cost	4.385 (2.610)
Log of leisure travelers	0.515 (0.063)
Year fixed effects	Included
Region fixed effects	Included
R-squared	0.8719
Number of observations	294

Note: Dependent variable is the log of the number of business travelers. Robust standard errors are in parentheses.

Table 5. Instrumental Variables Model of Exports

Variables	Unrestricted Model	Restricted Model
Log of GDP	0.518 (0.069)	Set at 1
Log of trade cost	-5.552 (1.878)	-2.029 (1.775)
Log of business travelers	0.460 (0.087)	0.383 (0.083)
Year fixed effects	Included	Included
Region fixed effects	Included	Included
R-squared	0.9674	0.9930
Number of observations	294	294

Note: Dependent variable is the log of U.S. exports to the region in the year. Instrumental variable is the log of the number of leisure travelers to the region in the year. Robust standard errors are in parentheses.

of business travelers, holding constant the other explanatory variables, increases U.S. commodity exports by approximately 2.84%.

However, our theoretical model indicates that the OLS estimates will be biased, since international business travel is jointly determined with U.S. commodity exports. To correct this bias, we use the number of leisure travelers to the region as an instrument for the number of U.S. business travelers to the region. First, we estimate the first-stage specification in (11) to confirm that the number of international leisure travelers to a region is correlated with the number of U.S. business travelers to the region. This is the first requirement of a valid instrumental variable. Table 4 reports the results of this first-stage estimation.

Next, we estimate the parameters of (10) in the second stage of the estimation. Table 5 reports the results. When we use the instrument, the estimated coefficient on the number of international business travelers in the unrestricted model increases to 0.46. This elasticity implies that a 10% increase in the number of business travelers increases the value of U.S. commodity exports by approximately 4.6%. We performed a Hausman test of whether OLS is an appropriate estimator of the parameters in (10). We reject the null hypothesis of the Hausman test (that the OLS estimator is consistent) in favor of the IV estimates.

Table 6. Alternative Instrumental Variable Model with a Lagged Effect of Business Travel

Variables	
Log of GDP	0.491 (0.075)
Log of trade cost	-4.363 (2.033)
Log of business travelers	0.447 (0.102)
In the prior year	
Year fixed effects	Included
Region fixed effects	Included
R-squared	0.9646
Number of observations	273

Note: Dependent variable is the log of U.S. exports to the region in the year. Instrumental variable is the log of the number of leisure travelers to the region in the prior year. Robust standard errors are in parentheses.

As a sensitivity analysis, we re-estimate the unrestricted model in Table 5 omitting China and then South America from the estimation sample. The purpose of this analysis is to test whether the estimated elasticity is sensitive to the inclusion of these two large destination regions, which could be outliers that drive the elasticity estimate. We find that the estimated elasticity is 0.436 when China is excluded and 0.435 when South America is excluded. In both cases, these elasticities are close to the elasticity reported in Table 5. The results indicate that the elasticity estimate is not sensitive to the set of countries included in the estimation sample.

We performed a second sensitivity analysis to provide further support for the validity of the instrumental variable. The second requirement of a valid instrumental variable is that it should not be correlated with the error term in the export equation. This issue is addressed in part by including the shipping cost measure in (10). The number of leisure travelers could be correlated with the cost of shipping commodities. By controlling for the cost of shipping goods, the specification in (10) ensures that the error term does not include shipping costs and is not likely to be correlated with the number of leisure travelers to the region.

Next, we adopt the Granger causality analysis of lagged correlation as a way to further reduce the likelihood of estimation bias. Specifically, we model exports in each year as a function of the number of business travelers in the prior year. In this case, the instrumental variable is the number of leisure travelers in the prior year. This should further reduce or eliminate correlation between the number of business travelers and the error term of the export equation. Table 6 reports this analysis. The elasticity estimate for this alternative specification is 0.447, which is still close to the estimate in Table 5.

As a third sensitivity analysis, we reestimate the model without the shipping cost measure. Although the shipping cost measure has a significant negative effect on U.S. commodity exports in the unrestricted model in Table 3, its omission from the specification does not have a significant effect on the estimated elasticity with respect to the number of international business travelers. Table 7 reports the additional econometric estimates.

Table 7. Instrumental Variables Model of Exports

Variables	Unrestricted Model	Unrestricted Model
Log of GDP	0.518 (0.069)	0.554 (0.067)
Log of trade cost	-5.552 (1.878)	Not included
Log of business travelers	0.460 (0.087)	0.466 (0.089)
Year fixed effects	Included	Included
Region fixed effects	Included	Included
R-squared	0.9674	0.9659
Number of observations	294	294

Note: Dependent variable is the log of U.S. exports to the region in the year. Instrumental variable is the log of the number of leisure travelers to the region in the year. Robust standard errors are in parentheses.

Estimation of the Effect on the Dollar Value of Exports

Finally, we use the econometric model in Table 5 to estimate the dollar value of the impact of international business travelers on U.S. commodity exports. Specifically, we calculate the marginal change in the value of the exports with respect to each additional U.S. business traveler, based on the unrestricted instrumental variables estimate of γ_j , the values of exports and the number of travelers reported for 2009, and the formula in (12).

$$\frac{dX_{jt}}{dB_{jt}} = \gamma_1 \frac{X_{jt}}{B_{jt}}. \quad (12)$$

This derivative quantifies the expected marginal export sales generated by each additional international business traveler. Table 8 reports the results for each of the 21 regions. In the first column of numbers, we report the dollar values of exports based on the unrestricted model. We estimate that a U.S. business traveler generates \$36,693 in additional U.S. exports on average across the countries. The values for China and the Netherlands (\$66,587 and \$69,590) are far above the average. The value for the United Kingdom (\$27,321) is below the average. The estimates based on the restricted model are similar.

The estimates can be further refined using a measure of the intensity of each international business trip, like the number of nights spent in the country. In the estimation of γ_j , the region and year fixed effects account for systematic differences in travel intensity between regions and between years. However, the estimated dollar values in Table 8 do not account for travel intensity. They represent the marginal contribution of each business traveler, not the marginal contribution of each travel day. To approximate the latter, we can divide the dollar-value estimates in Table 8 by 18.4, the average number of nights of U.S. overseas business trips in 2009.²

Conclusions

The results of our study support several conclusions. First, the elasticity of U.S. commodity exports with respect to

Table 8. Marginal Returns to Business Travel in 2009

Region	Unrestricted Model	Restricted Model
Africa	\$33,467	\$27,910
Australia	\$65,288	\$54,447
Caribbean	\$13,353	\$11,136
Central America	\$38,414	\$32,035
China	\$66,587	\$55,530
Eastern Europe	\$10,088	\$8,413
France	\$29,067	\$24,240
Germany	\$35,991	\$30,015
Hong Kong	\$27,232	\$22,711
India	\$26,176	\$21,830
Ireland	\$24,941	\$20,799
Italy	\$20,045	\$16,716
Japan	\$33,692	\$28,098
Korea	\$48,444	\$40,400
Middle East	\$40,991	\$34,184
Netherlands	\$69,596	\$58,035
South America	\$53,338	\$44,481
Spain	\$29,997	\$25,016
Switzerland	\$36,656	\$30,570
Taiwan	\$39,871	\$33,250
United Kingdom	\$27,321	\$22,784

international business travel is approximately 0.46. According to this estimate, a reduction in travel costs that increases international business travel to a country by 10% will generate a 4.6% increase in U.S. commodity exports to the country. Second, we find that the results are sensitive to the estimation technique. The IV estimate of the elasticity is almost twice the OLS estimate that ignores the joint endogeneity of business travel and trade flows.

Third, using U.S. export values reported for 2009, we can assign a dollar value to the impact of international business travel on exports. We estimate that each additional international business trip due to a reduction in international business costs will increase U.S. commodity exports to the visited country by \$36,693 per year on average. The impact of business travel on trade varies considerably across the 21 destination regions that we analyze.

The analysis of U.S. exports to the 21 regions has some general implications for the travel and tourism literature. The model provides a specific measure of the economic benefits generated by international business travel, and it provides validation for the theoretical framework discussed earlier. The article develops a statistical framework that can be applied to exports to a broader set of destination countries and a broader set of source countries, if the data are available.

Finally, the model has several policy implications. If policies that encourage infrastructure investments and reduce visa restrictions and other travel barriers can expand U.S. international business travel, then they will likely generate a significant expansion of U.S. commodity exports.

Acknowledgment

The views expressed are those of the authors and do not necessarily reflect those of the U.S. Department of Commerce, the Secretary of Commerce, the International Trade Administration, or the Undersecretary for International Trade. We thank Alis Asadurian, Joseph Flynn, Phillip McCalman, and two anonymous referees for helpful suggestions.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

Notes

1. For example, the models in Anderson and van Wincoop (2003) and Helpman, Melitz, and Rubinstein (2008) adopt this restriction.
2. The publicly available SIAT travel data do not provide detail on the number of nights by region, but they report this average value.

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